

《线性电路实验》实验报告

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1 实验目的

- (1) 理解运放主要参数的定义及对电路性能的影响, 增强对运放应用电路的理解;
- (2) 掌握运放主要参数的测量方法;
- (3) 测量 LM741CN, LM358, LM318N, OP177 等典型集成运放的基本参数

2 实验仪器

- (1) 数字万用表: Unit UT61E (C190241394)
- (2) 数字示波器: RIGOL 200MSO2202A (DS2F192200361)
- (3) 信号发生器: GWINSTEK AFG-22225 (GER910370)
- (4) 数字直流电源: GWINSTEK GPD-3303S (GES813705)
- (5) 多功能数字测量仪: Analog Discovery 1 (D704387)
- (6) 运放基本参数测试板: Basic Op Amp Measurement Board v2
- (7) 电压线性运算器: Voltage Linear Operation Board
- (8) 其它: 运算放大器 LM741CN, LM258P, LM318N, OP177 和 OPA197、面包板, 电容、电阻、杜邦线等

3 实验内容及实验结果

若无特别说明，后续实验均默认示波器为 dc coupling 模式。

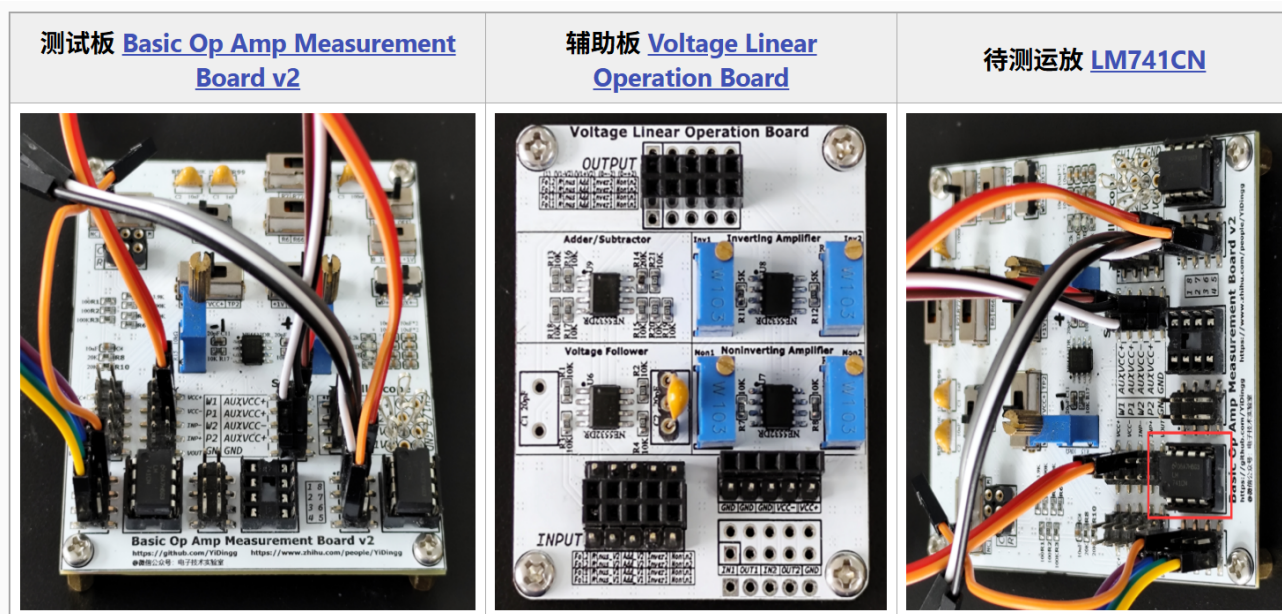


Figure 1: Using the basic op amp measurement board v2 and the linear operation board to measure the parameters of op amps

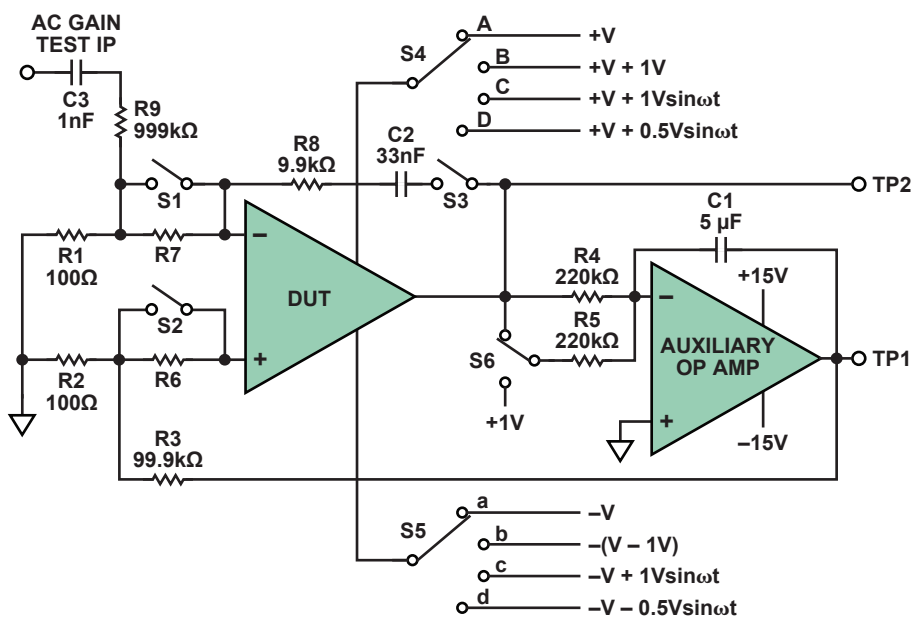


Figure 2: The reference measurement circuit from ADI: *Simple Op Amp Measurements* written by James M. Bryant

Refer to [YiDingg's Website > Electronic Blogs > Op Amp Measurement of LM741CN](#) for the detailed record of the experiment, the following shows the summary of the measurement results.

3.1 DC Measurement of LM741CN

The DC measurement results of LM741CN are shown in Table 1.

Table 1: DC measurement results of LM741CN

Parameter	Symbol	Value
Input offset voltage	V_{OS}	- 0.41398 mV
Non-inverting input bias current	I_{B+}	59.6609 nA
Inverting input bias current	I_{B-}	56.4179 nA
Input bias current	I_B	58.0394 nA
Input offset current	I_{OS}	1.6215 nA
DC small-signal open-loop gain	$A_{v,DC}$	+ 9978.4 V/mV (140.0 dB)
		- 11406.0 V/mV (141.1 dB)
		+ -3003.0 V/mV (129.6 dB)
DC common-mode rejection ratio	$CMRR_{DC}$	- 2929.8 V/mV (129.3 dB)
		-245.5 V/mV (107.8 dB)
DC power supply rejection ratio	$PSRR_{DC}$	-245.5 V/mV (107.8 dB)

3.2 AC Gain Measurement of LM741CN

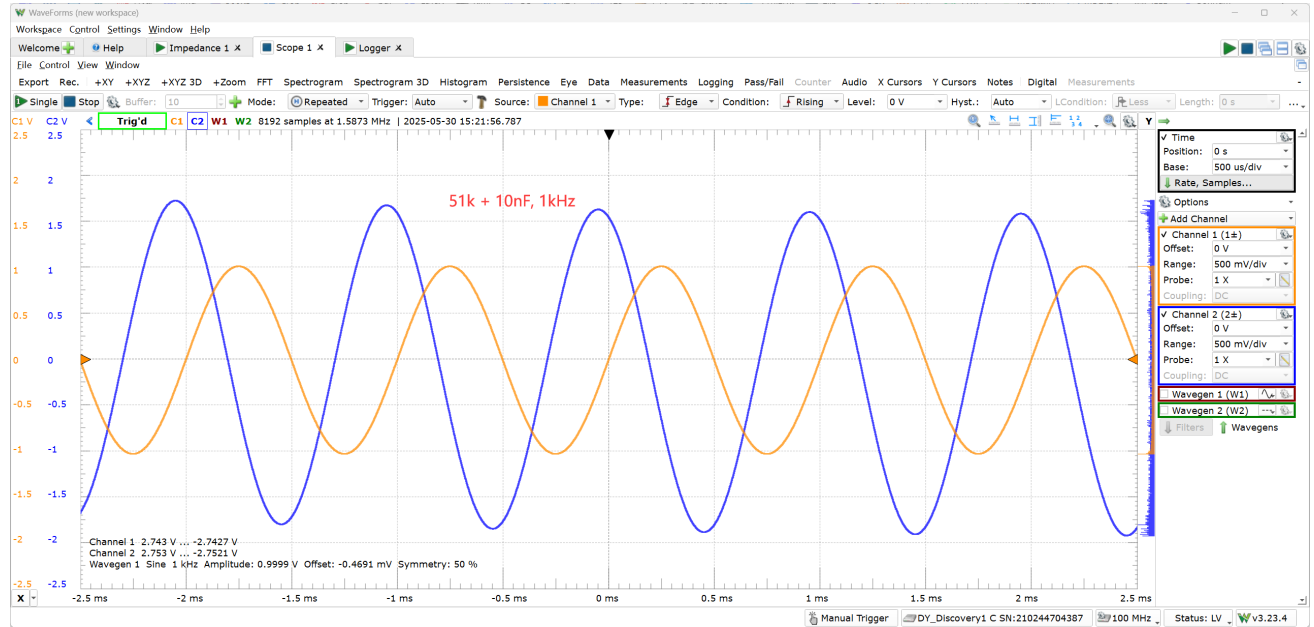


Figure 3: Input-output waveform in ac gain measurement of LM741CN at the frequency of 1 kHz; CH1 (orange) input (AC INPUT), CH2 (blue) output (TP2)

Neglecting the effect of the overall dc negative feedback (R_2 and R_3 in Figure 2), the corrected ac gain calculation formula is given by:

$$\begin{cases} v_{out} = v_{TP2} = A_v v_{eff} \\ v_{eff} = 0 - \frac{R_1}{R_1 + R_9 + \frac{1}{j\omega C_3}} \cdot v_{acin} \end{cases} \Rightarrow A_v = \left(1 + \frac{R_9}{R_1} + \frac{1}{j2\pi f R_1 C_3} \right) \times \left(-\frac{v_{TP2}}{v_{acin}} \right) \quad (1)$$

Note that the lowercase letters here denote the complex numbers with the magnitude and phase of the ac signals, i.e., $v_{TP2} = |v_{TP2}|e^{j\phi_{TP2}}$ and $v_{acin} = |v_{acin}|e^{j\phi_{acin}}$.

The complete calculation formula, which includes the effect of overall dc negative feedback, can be found in [YiDingg's Website > Electronic Blogs > The Correction of the AC Gain Equation in the ADI Op Amp Measurement Methods](#).

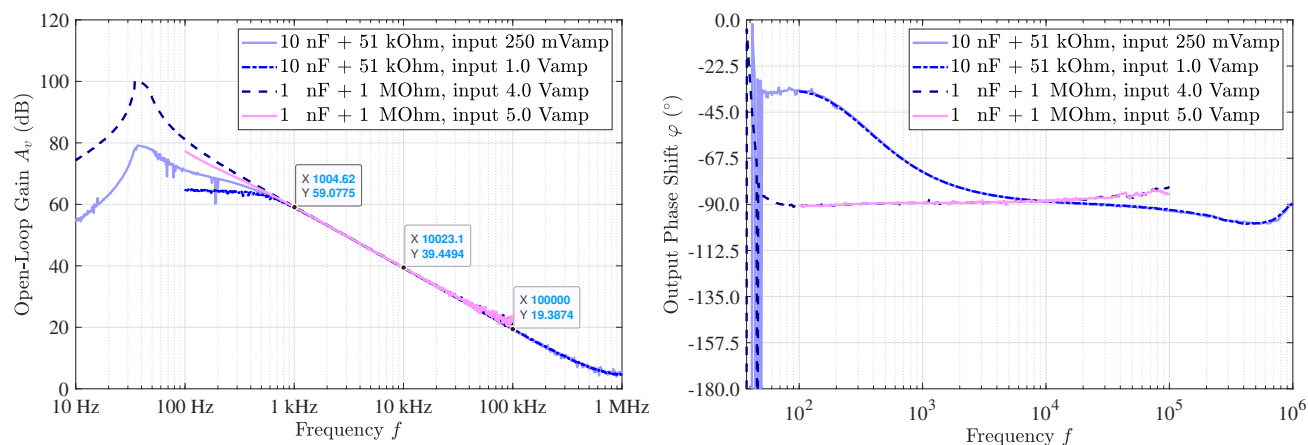


Figure 4: Frequency response measurement results of LM741CN

3.3 AC PSRR and CMRR Measurement of LM741CN

As shown below, we use the voltage linear operation board to generate the power supply for CMRR and PSRR measurements:

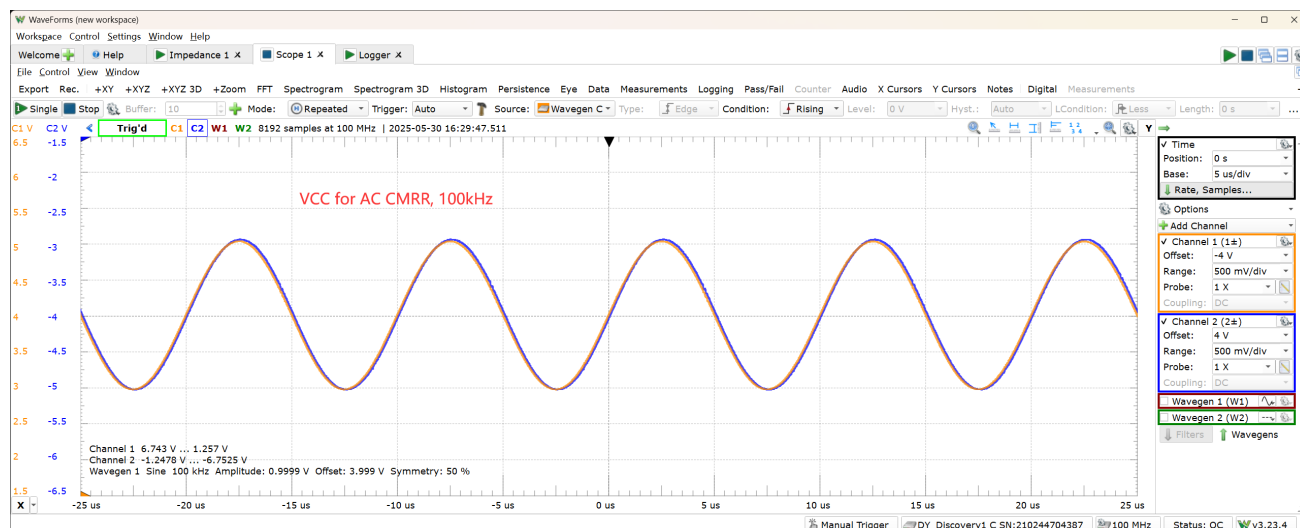


Figure 5: Generation of power supply for CMRR measurement; $V_{CC+} = 4\text{ V} + 1\text{ V} \times \sin \omega t$, $V_{CC-} = -4\text{ V} + 1\text{ V} \times \sin \omega t$

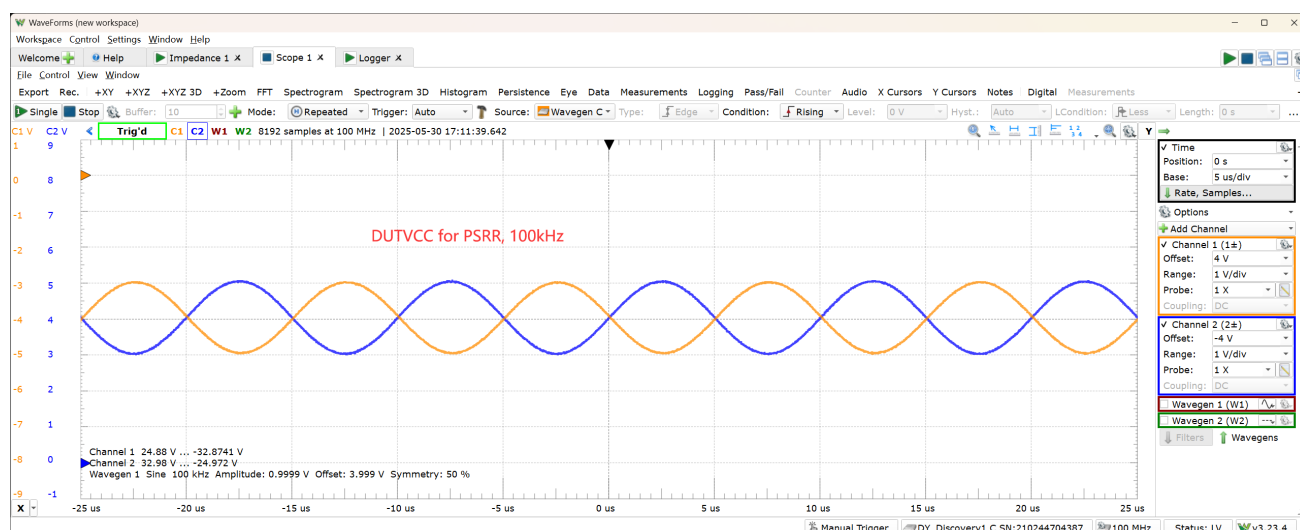


Figure 6: Generation of power supply for PSS measurement; $V_{CC+} = 4\text{ V} + 1\text{ V} \times \sin \omega t$, $V_{CC-} = -4\text{ V} - 1\text{ V} \times \sin \omega t$

The corrected CMRR and PSRR calculation formulas are given by:

$$\left\{ \begin{array}{l} \text{CMRR} = \frac{A_{DM-DM}}{A_{CM-DM}} = \frac{\frac{A_v \Delta V_{in,DM}}{\Delta V_{in,DM}}}{\frac{A_v \Delta V_{OS}}{\Delta V_{OS}}} = \frac{\Delta V_{CM}}{\Delta V_{OS}} \\ v_{W1} = v_{W2} \Rightarrow \Delta V_{CM} = v_{CM} = -v_{W1} \\ \Delta V_{OS} = v_{OS} = \frac{v_{out}}{1 + \frac{R_8}{R_1} + \frac{1}{j\omega R_1 C_2}} = \frac{v_{TP2}}{101 + \frac{10^5}{j\omega}} \end{array} \right. \Rightarrow \text{CMRR} = \left(101 + \frac{1 \times 10^5}{j 2\pi f} \right) \times \left(-\frac{v_{W1}}{v_{TP2}} \right) \quad (2)$$

$$\left\{ \begin{array}{l} \text{PSRR} = \frac{A_{DM-DM}}{A_{PS-DM}} = \frac{\frac{A_v \Delta V_{in,DM}}{\Delta V_{in,DM}}}{\frac{A_v \Delta V_{OS}}{\Delta V_{OS}}} = \frac{\Delta V_S}{\Delta V_{OS}} \\ v_{W1} = v_{W2} \Rightarrow \Delta V_S = v_S = 2 v_{W1} \\ \Delta V_{OS} = v_{OS} = \frac{v_{TP2}}{101 + \frac{10^5}{j\omega}} \end{array} \right. \Rightarrow \text{PSRR} = \left(101 + \frac{1 \times 10^5}{j 2\pi f} \right) \times \left(\frac{v_{W1}}{v_{TP2}} \right) \times 2 \quad (3)$$

Note that we used 10 kOhm for R8 and 100 Ohm for R1 hence the ratio is 101 instead of 100 here.

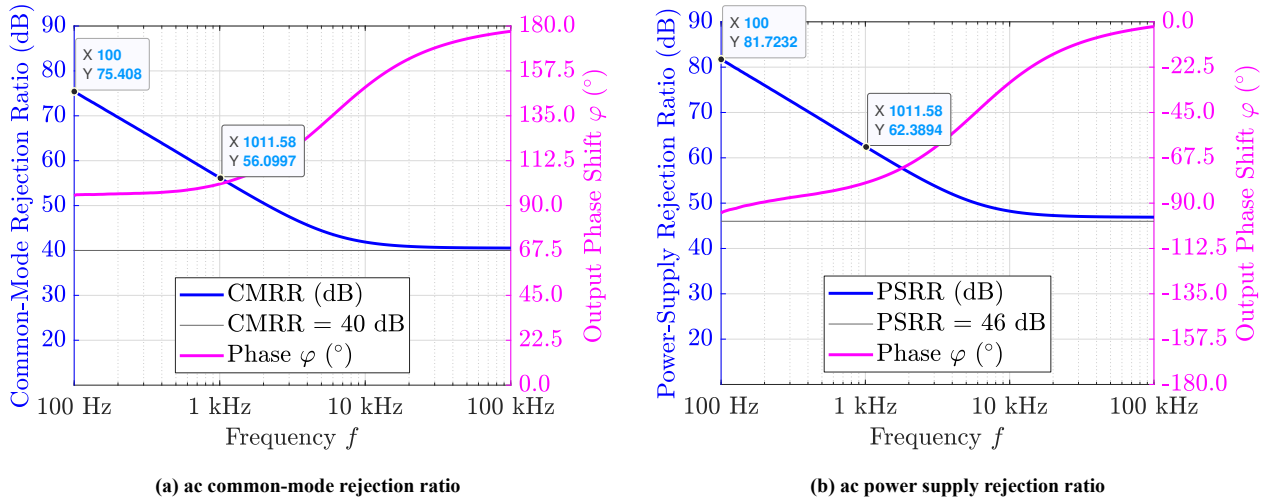


Figure 7: AC CMRR and PSRR measurement results of LM741CN

As can be seen from the above figure, the minimum measurable value of CMRR and PSRR is 40 dB and 46 dB, respectively. We'll discuss this phenomenon later.

3.4 Slew Rate Measurement of LM741CN

In the slew rate and full power bandwidth measurements, the op amp is configured as a simple unity buffer, i.e., the output terminal is directly connected to the inverting input terminal, and the input signal is injected into the non-inverting input terminal.

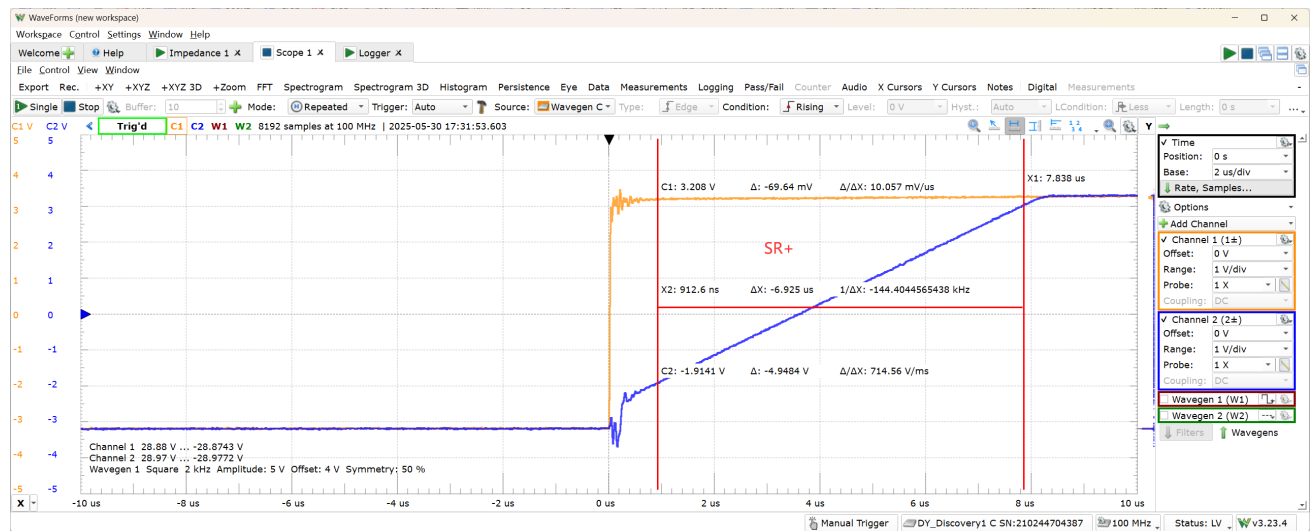


Figure 8: Rising slew rate waveform of LM741CN with 5 Vamp square wave input



Figure 9: Falling slew rate waveform of LM741CN with 5 Vamp square wave input

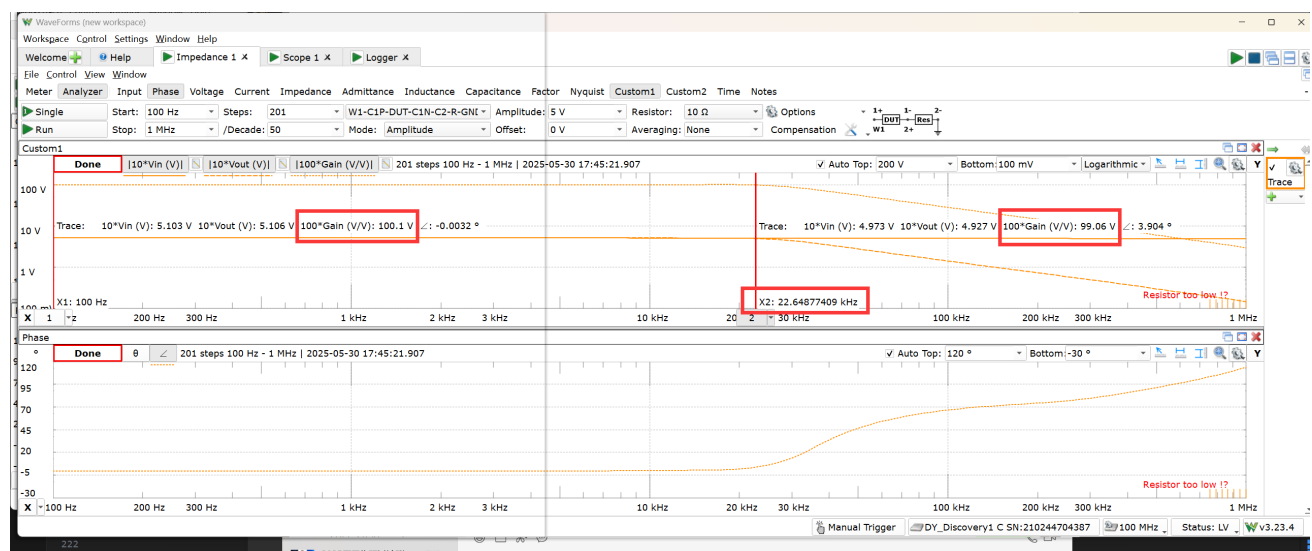


Figure 10: Output amplitude of LM318N with nominal 10 Vamp output as a function of frequency

Note that our definition of FPBW (full-power bandwidth) here is the frequency where the output amplitude drops by 1 % of its nominal value, instead of the definition under THD, i.e., the frequency where the THD of output signal reaches 1 %. The SR and FPBW results can be summarized in Table 2.

Table 2: Slew rate measurement results of LM741CN

Parameter	Symbol	Value
Rising slew rate	SR_+	+ 0.7146 V/us
Falling slew rate	SR_-	- 0.6099 V/us
Full power bandwidth	FPBW	22.6488 kHz @ 5 Vamp

3.5 Summary of Results

Table 3: DC measurement results of LM741CN

Parameter	Symbol	Value
Input offset voltage	V_{OS}	- 0.41398 mV
Non-inverting input bias current	I_{B+}	59.6609 nA
Inverting input bias current	I_{B-}	56.4179 nA
Input bias current	I_B	58.0394 nA
Input offset current	I_{OS}	1.6215 nA
DC small-signal open-loop gain	$A_{v,DC}$	+ 9978.4 V/mV (140.0 dB)
		- 11406.0 V/mV (141.1 dB)
		-3003.0 V/mV (129.6 dB)
DC common-mode rejection ratio	$CMRR_{DC}$	- 2929.8 V/mV (129.3 dB)
		-245.5 V/mV (107.8 dB)
DC power supply rejection ratio	$PSRR_{DC}$	-245.5 V/mV (107.8 dB)

Table 4: AC measurement results of LM741CN

AC Parameter	Symbol	Value
Gain-bandwidth product	GBW	0.9319 MHz
AC open-loop gain @ 1 kHz	$A_{v,1kHz}$	59.1 dB
AC open-loop gain @ 10 kHz	$A_{v,10kHz}$	39.4 dB
AC open-loop gain @ 100 kHz	$A_{v,100kHz}$	19.4 dB
AC common-mode rejection ratio	$CMRR_{100Hz}$	75.4 dB
AC common-mode rejection ratio	$CMRR_{1kHz}$	56.1 dB
AC power supply rejection ratio	$PSRR_{100Hz}$	81.7 dB
AC power supply rejection ratio	$PSRR_{1kHz}$	62.4 dB

Table 5: Slew rate measurement results of LM741CN

Parameter	Symbol	Value
Rising slew rate	SR_+	+ 0.7146 V/us
Falling slew rate	SR_-	- 0.6099 V/us
Full power bandwidth	FPBW	22.6488 kHz @ 5 Vamp

4 思考题

4.1 实验中有无遇到异常现象，如何解决？

4.1.1 TP2 电压存在漂移

测量输入失调电压和输入偏置电流时，输出节点 TP1 会以很慢的速率在一定范围内漂移 (将记录时间拉长至五分钟左右可轻易观察到)，波动幅度大约为平均值的 $\pm 10\%$ ，导致一定的测量误差。这是 ADI basic op amp measurement method 中的系统误差，可以通过调整辅助运放和全局反馈电阻以提高反馈稳定性，进而改善此确定，但是无法完全避免 (因为失调电压本身就会漂移)。

4.1.2 DC Gain 测量结果为负值

这是运放内部的“热反馈”造成的，具体原理见论文 [The monolithic op amp: a tutorial study](#)，全引是：

J. E. Solomon, “The monolithic op amp: a tutorial study”, in IEEE Journal of Solid-State Circuits, vol. 9, no. 6, pp. 314-332, Dec. 1974, doi: 10.1109/JSSC.1974.1050524.

4.1.3 AC Gain 低频测量结果明显不合理

这个问题我们在很久以前就发现了，但至今没有找到合理的解释。我对 AC Gain 的测量电路也作过完整、严谨的推导，但是最终却得到“不论辅助运放的性能如何，公式 (1) 在很宽的频率范围内都符合得很好”的结论，这与实验结果明显不符。不过，由于热反馈会影响 DC Gain 的测量，那么我们有理由猜测低频增益数据异常可能也是热反馈造成的，毕竟低频更接近 DC。具体原因为何，还有待进一步的研究。

4.1.4 AC CMRR 和 AC PSRR 存在“渐近线”

以共模抑制比为例，其渐近线出现在 $\text{CMRR} = 40.0864 \text{ dB} = 101 \text{ V/V}$ 处，这是因为当共模抑制比足够低时，运放的输出端 (TP2) 完全“follow”了输入共模的变化量，而我们的输入共模只有 1 V_{amp} ，因此输出端的变化量也不会超过 1 V_{amp} ，此时公式 (2) 退化为：

$$\text{CMRR} = \left(1 + \frac{R_8}{R_1} + \frac{1 \times 10^5}{j2\pi f}\right) \times \left(-\frac{v_{W1}}{v_{TP2}}\right) = 101 \times (-1) \text{ V/V} = -101 \text{ V/V} \quad (4)$$

也就是说，电阻 R_8 、 R_1 的值，以及输入共模幅度共同限制了 CMRR 的测量范围。因此，减小比值 $\frac{R_8}{R_1}$ ，或增大输入共模幅度 (也即增大 VCC 的交流幅度) 便可解决这个问题 (拓宽测量范围)。

同理，PSRR 的渐近线出现在 $\text{PSRR} = 46.1070 \text{ dB} = 202 \text{ V/V}$ 处，原因同上。

4.2 运放还有哪些重要参数？

除本实验所涉及的一些基本参数外，运放还有一些重要参数，如：

- (1) 小信号输出阻抗 (简称输出阻抗)
- (2) 小信号差模输入阻抗 (简称输入阻抗)
- (3) 大信号共模输入阻抗 (简称共模输入阻抗)
- (4) 输入输出寄生电容
- (5) 建立时间 (settling time)
- (6) 等效输入电压噪声谱密度 $\frac{dV_n^2}{df}$
- (7) 等效输入电流噪声谱密度 $\frac{dI_n^2}{df}$

附录 A 原始数据记录表

YiDingg's Website > Electronic Blogs > Op Amp Measurement of LM741CN

<https://yidingg.github.io/YiDingg/#!/Electronics/Op%20Amp%20Measurement%20of%20LM741CN>

附录 B Matlab Codes

```

1  LCE_08_OpAmpMeasurement
2
3  20250530 LM741CN DC 数据处理
4  %% 运放测量板测量运放参数
5  R_7 = 3.9e3;
6  R_77 = 390e3;
7  R_6 = 3.9e3;
8  R_66 = 390e3;
9
10 %% 实验数据处理
11
12 % 0. 示波器 dc 校准
13 V_CH1_0 = -6.154e-3;
14 V_CH2_0 = 7.863e-3;
15
16 % 1. Vos
17 Vos_TP1 = -420.55e-3;
18 Vos_TP2 = 19.709e-3;
19 Vos = (Vos_TP1 - V_CH1_0)/(1001);
20 disp(['Vos = ', num2str(Vos*1000), ' mV'])
21
22 % 2.1 Ib+ (R6)
23 Ib_pos_DeltaTP1 = (-187.59e-3) - (-420.5e-3);
24 Ib_pos_Resistor = R_6; % R_6 or R_66
25 Ib_pos = +(Ib_pos_DeltaTP1)/(1001*Ib_pos_Resistor);
26 disp(['Ib+ = ', num2str(Ib_pos*10^9), ' nA'])
27
28 % 2.2 Ib- (R7)
29 Ib_neg_DeltaTP1 = (-639.45e-3) - (-419.2e-3);
30 Ib_neg_Resistor = R_7; % R_7 or R_77
31 Ib_neg = -(Ib_neg_DeltaTP1)/(1001*Ib_neg_Resistor);
32 disp(['Ib- = ', num2str(Ib_neg*10^9), ' nA'])
33
34 Ib = 0.5*(Ib_pos + Ib_neg);
35 Ib_os = 0.5*(Ib_pos - Ib_neg);
36 disp(['Ib = ', num2str(Ib*10^9), ' nA'])
37 disp(['Ib_os = ', num2str(Ib_os*10^9), ' nA'])
38
39 % 3.1 DC Gain (1)
40 DC_Gain_1_DeltaTP1 = (-421.35e-3) - (Vos_TP1);
41 DC_Gain_1_DeltaTP2 = (-7.955) - (Vos_TP2);
42 Av_dc_1 = 1001*DC_Gain_1_DeltaTP2/DC_Gain_1_DeltaTP1;
43 disp(['DC Gain 1 = ', num2str(Av_dc_1, '%.2e'), ' = ', num2str(20*log(abs(Av_dc_1))/log(10)), ' dB'])
44
45 % 3.2 DC Gain (2)
46 DC_Gain_2_DeltaTP1 = (-419.65e-3) - (-420.35e-3);
47 DC_Gain_2_DeltaTP2 = (-7.956) - (20.194e-3);
48 Av_dc_2 = 1001*DC_Gain_2_DeltaTP2/DC_Gain_2_DeltaTP1;
49 disp(['DC Gain 2 = ', num2str(Av_dc_2, '%.2e'), ' = ', num2str(20*log(abs(Av_dc_2))/log(10)), ' dB'])
50
51 % DC PSRR
52 PSRR_DeltaVs = 2*(15 - 5);

```

```

53 PSRR_DeltaTP1 = (-445.7e-3) - (-364.14e-3);
54 PSRR = 1001*PSRR_DeltaVs/PSRR_DeltaTP1;
55 disp(['DC PSRR = ', num2str(PSRR), ' = ', num2str(20*log(abs(PSRR))/log(10)), ' dB'])
56
57 % DC CMRR (1)
58 CMRR_DeltaVcm = 0.5*(15 - 9) - 0.5*(12 - 12);
59 CMRR_DeltaTP1 = (-424.3e-3) - (-423.3e-3);
60 CMRR_1 = 1001*CMRR_DeltaVcm/CMRR_DeltaTP1;
61 disp(['DC CMRR = ', num2str(CMRR_1), ' = ', num2str(20*log(abs(CMRR_1))/log(10)), ' dB'])
62
63 % DC CMRR (2)
64 CMRR_DeltaVcm = 0.5*(15 - 9) - 0.5*(9 - 15);
65 CMRR_DeltaTP1 = (-425.05e-3) - (-423e-3);
66 CMRR_2 = 1001*CMRR_DeltaVcm/CMRR_DeltaTP1;
67 disp(['DC CMRR = ', num2str(CMRR_2), ' = ', num2str(20*log(abs(CMRR_2))/log(10)), ' dB'])
68
69 disp('-----')
70 disp('>>>----- LM741CN DC 参数汇总 -----<<<')
71 disp(['Vos = ', num2str(Vos*1000), ' mV'])
72 disp(['Ib+ = ', num2str(Ib_pos*10^9), ' nA'])
73 disp(['Ib- = ', num2str(Ib_neg*10^9), ' nA'])
74 disp(['Ib = ', num2str(Ib*10^9), ' nA'])
75 disp(['Ib_os = ', num2str(Ib_os*10^9), ' nA'])
76 disp(['DC Gain (1) = ', num2str(Av_dc_1/1000, '%.4f'), ' V/mV = ', num2str(20*log(abs(Av_dc_1))/log(10)),
77      ' dB'])
77 disp(['DC Gain (2) = ', num2str(Av_dc_2/1000, '%.4f'), ' V/mV = ', num2str(20*log(abs(Av_dc_2))/log(10)),
78      ' dB'])
78 disp(['DC PSRR = ', num2str(PSRR/1000, '%.4f'), ' V/mV = ', num2str(20*log(abs(PSRR))/log(10)), ' dB'])
79 disp(['DC CMRR (1) = ', num2str(CMRR_1/1000, '%.4f'), ' V/mV = ', num2str(20*log(abs(CMRR_1))/log(10)),
80      ' dB'])
80 disp(['DC CMRR (2) = ', num2str(CMRR_2/1000, '%.4f'), ' V/mV = ', num2str(20*log(abs(CMRR_2))/log(10)),
81      ' dB'])
81 disp('-----')
82 disp('-----')
83
84
85
86 20250530 LM741CN AC Gain 数据处理
87 clc, clear
88 data = readmatrix("D:\aa_MyExperimentData\Raw data backup\[op amp] ac gain, LM741CN, input 250 mVamp, 10
      nF + 51 kOhm, 10 Hz to 1 MHz.txt");
89 stc1 = MyDataProcessor_OpAmp_ACGain_10Hzto1MHz(data, 1);
90
91 data = readmatrix("D:\aa_MyExperimentData\Raw data backup\[op amp] ac gain, LM741CN, input 1 Vamp, 10 nF
      + 51 kOhm, 100 Hz to 1 MHz.txt");
92 stc2 = MyDataProcessor_OpAmp_ACGain_10Hzto1MHz(data, 1);
93 stc2.axes.XLim(1) = 10;
94
95 data = readmatrix("D:\aa_MyExperimentData\Raw data backup\[op amp] ac gain, LM741CN, input 5 Vamp, 1 nF
      + 1 MOhm, 10 Hz to 100 kHz.txt");
96 stc3 = MyDataProcessor_OpAmp_ACGain_10Hzto1MHz(data, 0);
97
98 data = readmatrix("D:\aa_MyExperimentData\Raw data backup\[op amp] ac gain, LM741CN, input 5 Vamp, 1 nF
      + 1 MOhm, 100 Hz to 100 kHz.txt");
99 stc4 = MyDataProcessor_OpAmp_ACGain_10Hzto1MHz(data, 0);
100
101 X_1 = [stc1.f; stc2.f; stc3.f; stc4.f];
102 Y_1 = [stc1.A_v_dB; stc2.A_v_dB; stc3.A_v_dB; stc4.A_v_dB];
103 X_2 = X_1;
104 Y_2 = [stc1.A_v_phase; stc2.A_v_phase; stc3.A_v_phase; stc4.A_v_phase];
105
106 stc = MyPlot_2window(X_1, Y_1, X_2, Y_2, 1);

```

```

107
108 % 调整图像属性
109 stc.ax1.XScale = 'log';
110 stc.ax1.XTick = logspace(1, 6, 6);
111 stc.ax1.XTickLabel = ["10 Hz", "100 Hz", "1 kHz", "10 kHz", "100 kHz", "1 MHz"];
112 stc.ax1.YLim = [0 120];
113 %xlim([2e2, 2e5])
114
115 stc.ax2.XScale = 'log';
116 stc.ax2.XTick = logspace(1, 6, 6);
117 stc.ax1.XTickLabel = ["10 Hz", "100 Hz", "1 kHz", "10 kHz", "100 kHz", "1 MHz"];
118 stc.ax2.YLim = [-180 0]; YTick = -180:22.5:0;
119 stc.ax2.YTick = YTick;
120 stc.ax2.YTickLabel = num2str(YTick, '%.1f');
121
122 stc.plot1.leg.String = [
123     "10 nF + 51 kOhm, input 250 mVamp"
124     "10 nF + 51 kOhm, input 1.0 Vamp"
125     "1\,\,\, nF + 1 \,MOhm, input 4.0 Vamp"
126     "1\,\,\, nF + 1 \,MOhm, input 5.0 Vamp"
127 ];
128 stc.plot1.leg.Location = 'northeast';
129 stc.plot2.leg.String = [
130     "10 nF + 51 kOhm, input 250 mVamp"
131     "10 nF + 51 kOhm, input 1.0 Vamp"
132     "1\,\,\, nF + 1 \,MOhm, input 4.0 Vamp"
133     "1\,\,\, nF + 1 \,MOhm, input 5.0 Vamp"
134 ];
135 stc.plot2.leg.Visible = 'on';
136 stc.plot2.leg.Location = 'northeast';
137
138 stc.plot1.label.x.String = 'Frequency $f$';
139 stc.plot1.label.y.String = 'Open-Loop Gain $A_v$ (dB)';
140 stc.plot2.label.x.String = 'Frequency $f$';
141 stc.plot2.label.y.String = 'Output Phase Shift $\varphi$ (^{\circ})$';
142 MyFigure_ChangeSize(512*[3, 1]);
143 %MyFigure_ChangeSize_2048x512
144 %stc.fig.WindowStyle = 'modal';
145 %MyExport_pdf
146
147
148 20250530 LM741CN CMRR PSRR 数据处理
149 clc, clear
150 data = readmatrix("D:\aa_MyExperimentData\Raw data backup\[op amp] LM741CN, CMRR, VCC = 4V + 1Vamp, 100
    Hz to 100 kHz.txt");
151 data(:, 4) = 0.5*data(:, 4);
152 stc1 = MyDataProcessor_OpAmp_ACCMRR_100Hzto100kHz(data);
153 %stc1.leg.String = ["CMRR (dB)", "CMRR = 40 dB", "Phase"];
154 %MyExport_pdf_modal
155
156 data = readmatrix("D:\aa_MyExperimentData\Raw data backup\[op amp] LM741CN, PSRR, VCC = 4V + 1Vamp, 100
    Hz to 100 kHz.txt");
157 data(:, 4) = 0.5*data(:, 4);
158 stc2 = MyDataProcessor_OpAmp_ACPSRR_100Hzto100kHz(data);
159 %hold(stc2.axes, 'on')
160 %MyExport_pdf_modal
161
162 LM741CN FPBW 数据处理
163 "D:\aa_MyExperimentData\Raw data backup\[op amp] LM741CN, Full Power Frequency, input 5Vamp, 100 Hz to
    100 kHz.txt"
164
165 % 运放测量板测量运放参数

```

```

166 R_7 = 3.9e3;
167 R_77 = 390e3;
168 R_6 = 3.9e3;
169 R_66 = 390e3;
170
171
172
173 TP1_Vos = -420.55e-3; % V_IO
174 TP2_Vos = 19.709e-3;
175
176 TP1_Ib_R7 = 3.3392;
177 TP1_Ib_R6 = 3.6214;
178
179 DCGain_1_TP1 = -421.35e-3;
180 DCGain_1_TP2 = -7.955;
181 DCGain_2_TP1 = -420.95e-3;
182 DCGain_2_TP2 = -7.955;
183 DCGain_mean_TP1 = 0.5*(DCGain_1_TP1 + DCGain_2_TP1);
184 DCGain_mean_TP2 = 0.5*(DCGain_1_TP2 + DCGain_2_TP2);
185
186
187
188 TP1_PSRR = 3.3278;
189 Delta_Vs = 2*(16 - 12);
190
191
192 I_B_neg = -(TP1_Ib_R7 - TP1_Vos)/(R_7*1001);
193 I_B_neg_2 = -(3.4266 - 3.5654)/(R_7*1001);
194 I_B_pos = (TP1_Ib_R6 - TP1_Vos)/(R_6*1001);
195 I_B_pos_2 = (3.72 - 3.5566)/(R_6*1001);
196 A_v_dc = 1001 * (DCGain_1_TP2 - TP2_Vos) / (DCGain_1_TP1 - TP1_Vos);
197 A_v_dc_2 = 1001 * (DCGain_2_TP2 - TP2_Vos) / (DCGain_2_TP1 - TP1_Vos);
198 A_v_dc_mean = 1001 * (DCGain_mean_TP2 - TP2_Vos) / (DCGain_mean_TP1 - TP1_Vos);
199
200 PSRR_dc = abs( 1001 * Delta_Vs / (TP1_PSRR - TP1_Vos) );
201
202
203 disp(['V_IO = ', num2str(TP1_Vos/1001 * 1000), ' mV'])
204 disp(['I_B_neg = ', num2str(I_B_neg*10^9), ' nA'])
205 disp(['I_B_neg_2 = ', num2str(I_B_neg_2*10^9), ' nA'])
206 disp(['I_B_pos = ', num2str(I_B_pos*10^9), ' nA'])
207 disp(['I_B_pos_2 = ', num2str(I_B_pos_2*10^9), ' nA'])
208 I_B = 0.5*(I_B_pos_2 + I_B_neg_2)*10^9
209 I_OS = 0.5*(I_B_pos_2 - I_B_neg_2)*10^9
210
211 disp(['DC Gain 1 = ', num2str(A_v_dc, '%.2e'), ' = ', num2str(20*log(abs(A_v_dc))/log(10)), ' dB'])
212 disp(['DC Gain 2 = ', num2str(A_v_dc_2, '%.2e'), ' = ', num2str(20*log(abs(A_v_dc_2))/log(10)), ' dB'])
213 disp(['DC Gain mean = ', num2str(A_v_dc_mean, '%.2e'), ' = ', num2str(20*log(abs(A_v_dc_mean))/log(10)),
    ' dB'])
214 disp(['DC PSRR = ', num2str(PSRR_dc), ' = ', num2str(20*log(PSRR_dc)/log(10)), ' dB'])
215
216 GBWP = 2.051e3* 199.4;
217 disp(['GBWP = ', num2str(GBWP/10^6), ' MHz'])
218
219
220 20250528 ADI op amp 测量方法修正
221 syms v_out1 A_1 v_out2 R_2 R_3 v_1 omega R_4 A_2 C_1 R_1 R_9 C_3 v_in s
222
223 % 代入具体数值
224 if 0
225 C_3 = 10e-9;
226 R_9 = 51e3;

```

```

227 R_1 = 100;
228 R_2 = 100;
229 R_3 = 100e3;
230 R_4 = MyParallel(20, 20);
231 C_1 = 10e-6;
232 end
233
234 eq1 = v_1 == R_1 / (R_1 + R_9 + 1/(s*C_3)) * v_in
235 eq2 = v_out1 == A_1 * ( v_out2/(1+R_3/R_2) - v_1 )
236 eq3 = (v_out1 + v_out2/A_2) / R_4 == (-v_out2/A_2 - v_out2) / (1/s*C_1)
237 re_v_out2 = solve(eq2, v_out2)
238 eq3 = subs(eq3, v_out2, re_v_out2);
239 re_v_out1 = solve(eq3, v_out1);
240 re_v_out1 = simplifyFraction(re_v_out1)
241 re_v_out1_dividedBy_v_1 = simplify(re_v_out1/v_1)
242
243 limit(re_v_out1_dividedBy_v_1, A_2, inf)
244 simplify(subs(re_v_out1_dividedBy_v_1, A_2, A_1))
245
246
247 20250528 数据处理代码测试
248 %% AC Gain 数据处理
249
250 % 导入数据
251 clc, clear, close all
252 data = readmatrix("D:\aa_MyExperimentData\Raw data backup\[op amp] ac gain, discrete uA741, input 1 Vamp
    , 10 nF + 51 kOhm.txt");
253 V_in_abs = data(:, 4)'; % AC INPUT, 为参考相位
254 V_out_abs = data(:, 5)'; % TP2
255 f = data(:, 1)';
256 InputPhase = data(:, 2)';
257 V_out_phase = - InputPhase;
258
259 % 先对数据后半段重点滤波
260 index = 360:length(f);
261 window = 20;
262 InputPhase(index) = MyFilter_mean(InputPhase(index), window);
263 V_in_abs(index) = MyFilter_mean(V_in_abs(index), window);
264 V_out_phase(index) = MyFilter_mean(V_out_phase(index), window);
265
266
267 % 再对数据整体进行滤波
268 window = 3;
269 InputPhase = MyFilter_mean(InputPhase, window);
270 V_in_abs = MyFilter_mean(V_in_abs, window);
271 V_out_phase = MyFilter_mean(V_out_phase, window);
272
273 % 选择所使用的电容和电阻
274 if 1 % 51 kOhm + 10 nF
275     R_9 = 51e3;
276     C_3 = 10e-8;
277 end
278 if 0 % 1 MOhm + 1 nF
279     R_9 = 1e6;
280     C_3 = 1e-9;
281 end
282
283 % 将模长和相位转化为复数表达
284 R_1 = 100;
285 v_acin = V_in_abs;
286 v_TP2 = V_out_abs .* (cosd(V_out_phase) + 1j*sind(V_out_phase));
287 %v_TP2'

```

```

288 %MyArcTheta_complex_rad(v_TP2)'
289 %test = atand(real(v_TP2)./imag(v_TP2)) - 180
290
291 % 作出 A_v 波特图
292 A_v = ( 1 + R_9/R_1 + 1./(1j*2*pi.*f*R_1*C_3) ) .* (- v_TP2./v_acin);
293 A_v_abs = abs(A_v);
294 A_v_dB = 20*log(A_v_abs)/log(10);
295 A_v_phase = MyArcTheta_complex_deg(A_v) - 360;
296 stc = MyYYPlot(f, f, A_v_dB, A_v_phase);
297 stc.axes.XScale = 'log';
298
299 % 调整图像属性
300 ylim([0 80])
301 xlim([2e2, 2e5])
302 stc.axes.XTick = [2e2, 1e3, 2e3, 1e4, 2e4, 1e5, 2e5];
303 stc.axes.XTickLabel = ["200 Hz", "1 kHz", "2 kHz", "10 kHz", "20 kHz", "100 kHz", "200 kHz"];
304 yyaxis('right');
305 ylim([-180 0])
306 YTick = -180:22.5:0;
307 stc.axes.YTick = YTick;
308 stc.axes.YTickLabel = num2str(YTick, '%.1f');
309 stc.axes.XScale = 'log';
310 stc.legend.String = ["Gain $A_v$ (dB)"; "Phase $\varphi$ (^\circ)$"];
311 stc.label.x.String = 'Frequency $f$';
312 stc.label.y_left.String = 'Open-Loop Gain $A_v$ (dB)';
313 stc.label.y_right.String = 'Output Phase Shift $\varphi$ (^\circ)$';
314
315 % 导出图像
316 %MyFigure_ChangeSize_2048x512
317 %MyExport_pdf
318
319
320 A_v = (1+(1./(2*pi*f*10^(-8))+51000)/100) .* (V_out_abs./V_in_abs);
321 A_v_dB = 20*log(A_v)/log(10);
322
323 stc = MyYYPlot(f, f, A_v_dB, test);
324 stc.axes.XScale = 'log';
325 stc.legend.String = ["Gain $A_v$ (dB)"; "Phase $\varphi$ (^\circ)$"];
326 stc.label.x.String = 'Frequency $f$';
327 stc.label.y_left.String = 'Open-Loop Small-Signal Gain $A_v$ (dB)';
328 stc.label.y_right.String = 'Output Phase Shift $\varphi$ (^\circ)$';
329 MyFigure_ChangeSize_2048x512
330 ylim([0 80])
331 xlim([2e2, 2e5])
332 stc.axes.XTick = [2e2, 1e3, 2e3, 1e4, 2e4, 1e5, 2e5]
333 stc.axes.XTickLabel = ["200 Hz", "1 kHz", "2 kHz", "10 kHz", "20 kHz", "100 kHz", "200 kHz"];
334 yyaxis('right')
335 ylim([0 180])
336 YTick = 0:22.5:180;
337 stc.axes.YTick = YTick
338 stc.axes.YTickLabel = num2str(YTick, '%.1f');
339
340 MyPlot(f, A_v.*f)

```

附录 C 实验预习报告

《线性电路实验》预习报告

实验名称: 运放参数测量 指导教师: 王东雷 df4dac@sina.com
姓名: 丁毅 学号: 2023K8009908031 班级/专业: 2308/电子信息 分组序号: 2-06
实验日期: 2025.05.30 实验地点: 教学楼 607 是否调课/补课: 否 成绩: _____

1 实验目的

- (1) 理解运放主要参数的定义及对电路性能的影响, 增强对运放应用电路的理解;
- (2) 掌握运放主要参数的测量方法;
- (3) 测量 uA741, LM358, LM318N, OP177 等典型集成运放的基本参数

2 实验仪器

- (1) 数字万用表: Unit UT61E (C190241394)
- (2) 数字示波器: RIGOL 200MSO2202A (DS2F192200361)
- (3) 信号发生器: GWINSTEK AFG-22225 (GER910370)
- (4) 数字直流电源: GWINSTEK GPD-3303S (GES813705)
- (5) 多功能数字测量仪: **Analog Discovery 1** (D704387)
- (6) 运放基本参数测试板: **Basic Op Amp Measurement Board v2**
- (7) 电压线性运算器: **Voltage Linear Operation Board**
- (8) 其它: 运算放大器 uA741, LM258P, LM318N, OP177 和 OPA197、面包板, 电容、电阻、杜邦线等

3 实验内容概要

注: 我们并不按照讲义上给出的具体步骤进行实验, 所有数据均以图片形式先记录在个人网站中, 再统一转到 MATLAB 进行处理, 因此不需要纸质数据表格。

- (1) 测量 uA741 的全部基本参数, 包括 V_{OS} , $I_{B\pm}$, DC 和 AC 下的 A_v , PSRR, CMRR, slew rate 和 FPBW
- (2) 测量 LM258P 的 V_{OS} , $I_{B\pm}$
- (3) 测量 OP177 的 V_{OS} , $I_{B\pm}$ 和 $A_{v,DC}$
- (4) 测量 LM318N 的 slew rate, 以及在 10 Vamp output 下的输出摆幅曲线 (理论带宽 $f_p = \frac{SR}{2\pi V_{out,amp}}$)
- (5) (选做) 测量 OPA197 的各项参数, 包括 V_{OS} , $I_{B\pm}$, DC 和 AC 下的 A_v , PSRR, CMRR 以及 slew rate

4 uA741 原理图

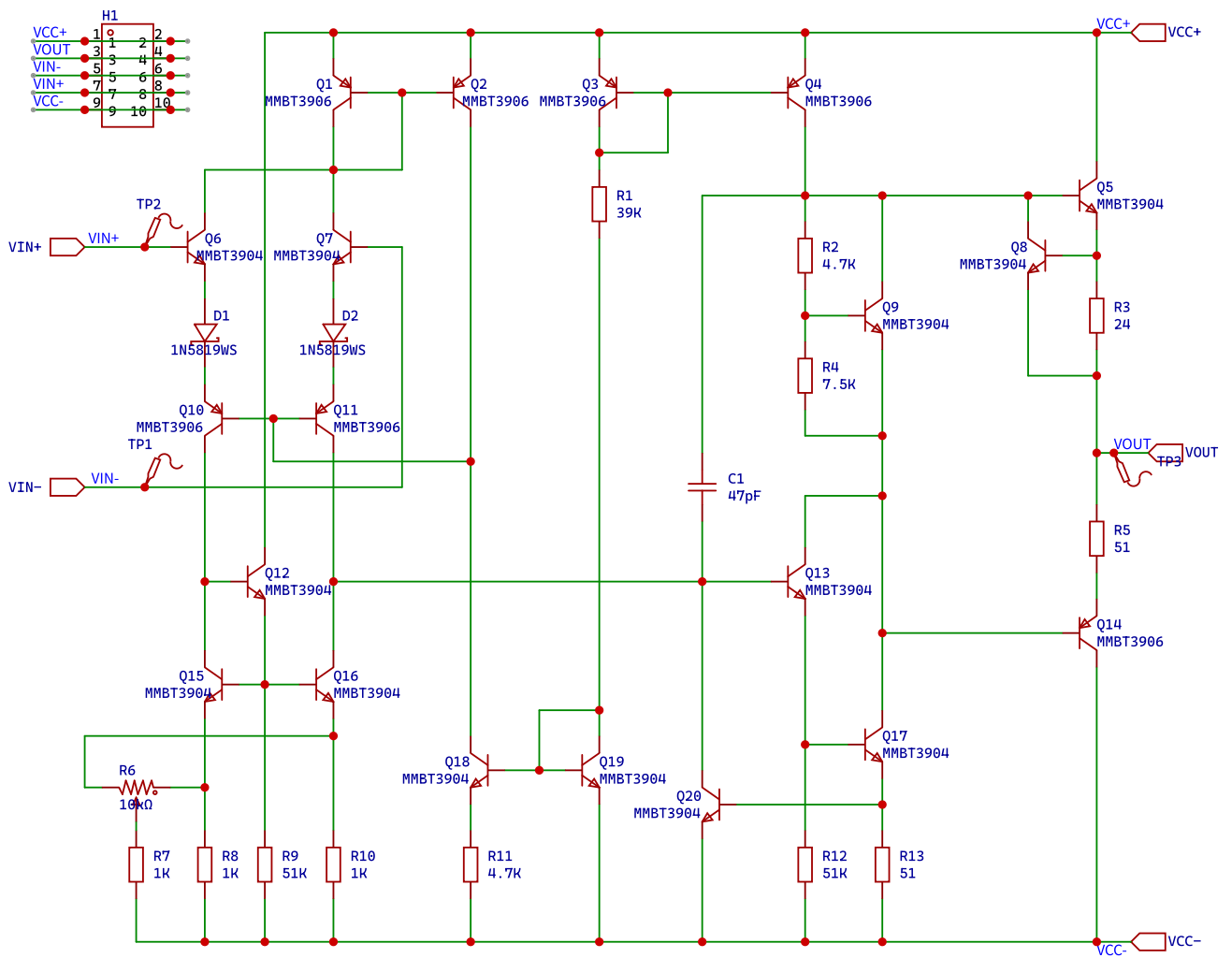


Figure 1: Circuit schematic of uA741